

Uncertainty-Aware Language-Grounded 3D Scene Graphs from Sparse RGB Views

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Abstract

We present a sparse-view pipeline for constructing language-grounded 3D scene graphs from casually captured RGB images. Feed-forward geometry models can now predict cameras, depth, and world-point maps without per-scene optimization, while 2D foundation models provide class-agnostic masks and open-vocabulary descriptors. However, these outputs remain view-local: they do not directly identify persistent objects, expose uncertain geometric evidence, or produce relation-bearing scene structure. Our method bridges this gap by lifting SAM proposals into VGGT world-point maps, attaching CLIP and DINOv2 descriptors, scoring lifted candidates with confidence and compactness uncertainty, fusing cross-view candidates into object nodes, and exporting open-vocabulary labels and spatial relations. The graph representation keeps symbolic predictions together with source views, proposal support, 3D point samples, and uncertainty values, making the output inspectable and reviewable. We evaluate the pipeline on a five-sequence TUM RGB-D sparse-view benchmark with 20 runs across 3, 5, 8, and 10 input views. Averaged over scenes, moving from 3 to 10 views increases lifted proposals from 54.0 to 173.6, fused objects from 31.8 to 72.2, multi-view objects from 11.8 to 28.2, and inferred relations from 585.2 to 2104.0. Against checked references initialized from the 10-view graphs, non-reference object-label F1 improves from 0.622 to 0.914 and relation-triplet F1 from 0.434 to 0.799 between 3 and 8 views. These reference-consistency scores measure sparse-view recovery rather than dense-scene recognition accuracy, establishing a reproducible graph construction baseline and a concrete path from predicted graphs to object-relation review.

Keywords: 3D scene graphs, sparse views, open vocabulary, 3D geometry, uncertainty

1. Introduction

Reconstructing a scene and explaining a scene are related but different goals. Modern geometry pipelines can recover camera motion, depth, points, surfaces, or radiance fields, yet downstream applications often require object-level statements: which pixels across views depict the same object, where that object is in 3D, and how it is spatially related to other objects. A robot or mixed-reality system, for example, benefits more from a statement such as “the monitor is on the desk” than from an unlabeled collection of points. This need is especially acute when the input is small: a few casual RGB views rather than a calibrated RGB-D stream, dense video, or already optimized map.

Classical structure-from-motion and SLAM systems remain powerful when overlap, calibration, and optimization time are available [1, 2]. Recent feed-forward geometry models have changed the starting point by predicting camera parameters, depth, and point maps directly from one or more images [3, 4, 5]. At the same time, 2D foundation models provide strong class-agnostic masks, language-aligned embeddings, and robust self-supervised descriptors through SAM, CLIP, and DINOv2 [6, 7, 8]. These components make sparse-view semantic 3D reasoning plausible, but they do not by themselves define an object-level graph. Masks are local to individual images, repeated instances may have similar descriptors, and feed-forward geometry is least reliable near occlusions, thin structures, textureless regions, and large viewpoint changes.

The key observation in this work is that sparse-view scene graph construction can be posed as evidence-preserving fusion over lifted 2D proposals. Rather than optimizing a new scene representation, we use VGGT world-point maps as a 2D-to-3D lookup table, lift SAM masks into candidate 3D object fragments, attach CLIP and DINOv2 descriptors, and merge compatible candidates across views. Each fused node retains its source proposals, view support, point samples, feature descriptor, open-vocabulary label, and uncertainty. Spatial predicates are then inferred over the fused object set. This design keeps the expensive perception models frozen while making the graph reasoning stage explicit, auditable, and easy to ablate.

The resulting setting differs from open-vocabulary 3D graph systems that assume posed RGB-D video, a dense point cloud, or a prebuilt map [9].

Our target is a lower-input regime: sparse RGB views processed by feed-forward geometry. We evaluate the system on five public TUM RGB-D sequences under 3/5/8/10-view protocols, report structural graph statistics for all 20 runs, and evaluate sparse-view recovery against checked references initialized from each 10-view graph. The references are not a substitute for independently authored dense ground truth, but they provide a reproducible measurement path for object labels and relation triplets.

Our contributions are:

- a sparse-RGB 3D scene graph pipeline that combines VGGT geometry, SAM proposals, and CLIP+DINOv2 proposal descriptors without per-scene optimization;
- an uncertainty-aware lifting representation that stores confidence, compactness, view support, source proposals, and point evidence for each object candidate;
- a cross-view object fusion and spatial-relation inference procedure that produces open-vocabulary object-relation graphs; and
- a five-sequence TUM RGB-D sparse-view protocol with generated tables, qualitative figures, annotation packets, and checked reference-consistency metrics [10].

2. Related Work

Feed-forward sparse-view geometry. Multi-view reconstruction and SLAM classically rely on correspondence, triangulation, bundle adjustment, and calibrated or densely overlapping observations. COLMAP remains a standard reference for incremental structure from motion [1], while ORB-SLAM3 demonstrates the strength of optimized visual and visual-inertial SLAM for coherent streams [2]. Feed-forward 3D models reduce this dependence on per-scene optimization. DUS3R casts reconstruction as point-map prediction from unconstrained images [3]; MAST3R adds 3D-aware dense matching [4]; and VGGT predicts camera parameters, depth, point maps, confidence, and tracks from one or many views [5]. These models provide the geometric substrate for our work. We use VGGT as a frozen source of pixel-aligned 3D evidence, but our output is not a reconstruction: it is an object-relation graph grounded in that geometry.

Open-vocabulary 2D and 3D semantics. Our 2D proposal and feature stack builds on SAM for class-agnostic masks, CLIP for language-aligned visual embeddings, and DINOv2 for robust self-supervised appearance descriptors [6, 7, 8]. In 3D, open-vocabulary scene understanding has been explored by lifting, distilling, or optimizing foundation-model features into point clouds, maps, masks, and neural scene representations. NeRF introduced continuous radiance fields for view synthesis [11]; LERF embeds language features into radiance fields for open-ended 3D queries [12]; and 3D Gaussian Splatting has become a common explicit substrate for fast scene representations [13]. OpenScene learns open-vocabulary point features aligned with image and text embeddings [14]; ConceptFusion builds queryable 3D maps by fusing pixel-aligned open-set features [15]; OpenMask3D aggregates CLIP features over class-agnostic 3D instance masks [16]; CUA-O3D studies cross-modal and uncertainty-aware agglomeration for open-vocabulary 3D scene understanding [17]; and OpenSplat3D performs open-vocabulary 3D instance segmentation in a Gaussian Splatting representation [18]. These methods motivate foundation-feature fusion in 3D. Our contribution is complementary: we construct explicit object-relation graphs from sparse RGB views without assuming a completed 3D map or optimizing a per-scene radiance field.

3D scene graphs. 3D scene graphs organize object instances and their relationships as structured representations for reasoning about scenes. Early work proposed unified graphs spanning objects, spaces, cameras, and semantic or spatial relationships [19]. 3DSSG introduced a large-scale dataset and learned prediction framework for semantic scene graphs from 3D indoor reconstructions [20]. RGB-D resources such as ScanNet further show the value of richly annotated reconstructed scenes for indoor understanding [21]. ConceptGraphs is closest to our output space: it builds open-vocabulary 3D scene graphs from posed RGB-D sequences by fusing 2D foundation-model outputs into object nodes and relations [9]. We share the object-relation motivation, but relax the input assumption. Instead of requiring a depth stream, posed RGB-D map, or completed 3D reconstruction, we lift sparse-view 2D proposals through feed-forward VGGT point maps and build object identities through cross-view geometric and semantic fusion.

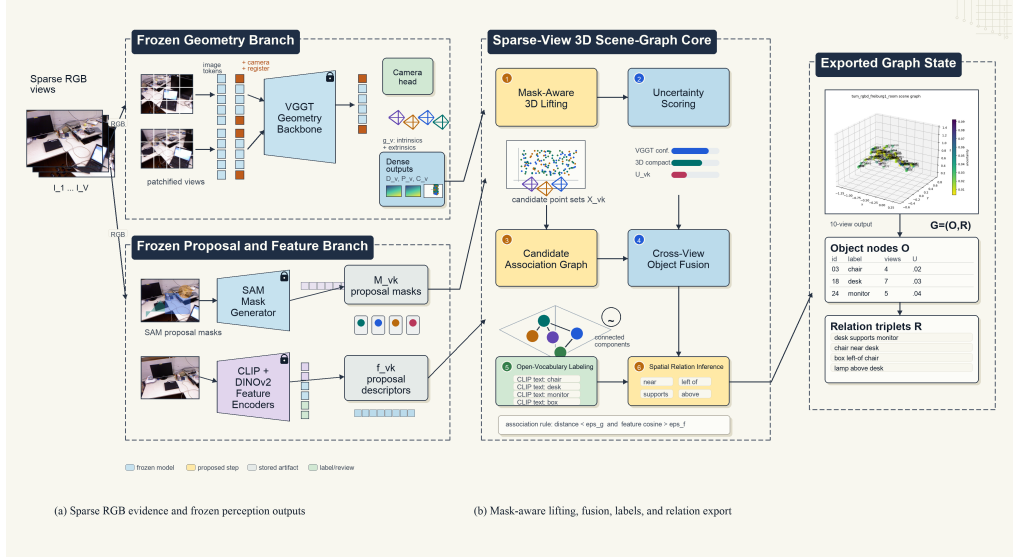


Figure 1: Architecture of the proposed sparse-view 3D scene graph system. The figure shows the actual artifacts passed between stages: sparse RGB frames, patch and camera tokens, frozen VGGT camera/depth/world-point/confidence outputs, SAM masks, CLIP+DINOv2 proposal descriptors, lifted 3D candidate point sets, uncertainty scores, association edges, fused objects, open-vocabulary labels, relation predicates, and the exported scene graph $\mathcal{G} = (\mathcal{O}, \mathcal{R})$.

3. Method

3.1. Problem Setup

Given a sparse set of RGB views $\mathcal{I} = \{I_1, \dots, I_V\}$, the goal is to produce a 3D scene graph $\mathcal{G} = (\mathcal{O}, \mathcal{R})$. Each object node $o_i \in \mathcal{O}$ stores a 3D point set, centroid, open-vocabulary label, view support, feature descriptor, source proposals, and uncertainty. Each relation edge $r_{ij} \in \mathcal{R}$ stores a spatial predicate between two object nodes. The method does not assume depth images, externally supplied camera poses, or a prebuilt mesh. It instead derives geometry, object proposals, and proposal descriptors from frozen feed-forward models, then performs object-level fusion and relation inference over the resulting evidence. Figure 1 summarizes the complete data flow.

3.2. Overview

The pipeline has five stages. First, VGGT predicts camera parameters, depth maps, world-point maps, and confidence maps for the selected

sparse views. Second, SAM generates view-local object proposals, and CLIP+DINOv2 descriptors are extracted for each proposal. Third, every proposal mask is lifted into the VGGT world-point map to form a 3D candidate object fragment with a confidence and compactness uncertainty score. Fourth, candidates from different views are connected when they are close in 3D and sufficiently similar in descriptor space; connected components become fused object nodes. Fifth, fused nodes are assigned open-vocabulary labels and connected by spatial relation predicates. The output graph keeps the predicted labels and relations together with the evidence used to produce them, so the graph can be inspected or converted into annotation review packets.

3.3. Sparse-View Geometry

We run VGGT on the selected sparse views and save camera intrinsics, extrinsics, depth maps, world-point maps, and confidence maps [5]. The world-point map for each view acts as a 2D-to-3D lookup table. This is the key simplification: object proposals can be lifted directly from image coordinates into a common 3D frame without first reconstructing a mesh, running bundle adjustment, or optimizing a radiance-field or Gaussian Splatting representation [1, 11, 13]. For view v , we denote the world-point map by $P_v(u) \in \mathbb{R}^3$ and the corresponding confidence map by $C_v(u)$ at image coordinate u .

3.4. 2D Proposals and Features

SAM produces class-agnostic masks for each view [6]. For proposal k in view v , we store the bounding box, confidence score, RLE mask, and visual descriptors. CLIP supplies the language-aligned component, while DINOv2 adds a complementary self-supervised appearance signal [7, 8]. When both descriptors are available, the normalized vectors are concatenated and renormalized:

$$f_{v,k} = \frac{[\hat{f}_{v,k}^{\text{CLIP}}, \hat{f}_{v,k}^{\text{DINO}}]}{\|[\hat{f}_{v,k}^{\text{CLIP}}, \hat{f}_{v,k}^{\text{DINO}}]\|_2}. \quad (1)$$

If one descriptor is unavailable, the valid normalized descriptor is retained. This keeps the fusion stage usable across proposal files with different feature backends.

3.5. Mask-Aware 3D Lifting

Each 2D proposal mask $M_{v,k}$ is resized to the VGGT geometry resolution. The mask selects world points from the corresponding view, while non-finite points and points below the confidence threshold τ_c are discarded:

$$X_{v,k} = \{P_v(u) \mid u \in M_{v,k}, C_v(u) \geq \tau_c\}. \quad (2)$$

Proposals with no valid lifted points are rejected. The remaining candidate node stores a sampled 3D point set $X_{v,k}$, centroid $\mu_{v,k}$, 3D bounding box, proposal metadata, visual feature $f_{v,k}$, and an uncertainty score. The uncertainty combines VGGT point confidence with the spatial compactness of the lifted 3D points:

$$U_{v,k} = \lambda(1 - \bar{C}_{v,k}) + (1 - \lambda) \text{clip}\left(\frac{\sigma(X_{v,k})}{s_0}, 0, 1\right), \quad (3)$$

where $\bar{C}_{v,k}$ is the mean confidence of the selected points and $\sigma(X_{v,k})$ summarizes spread around the median 3D point. In the implementation, $\lambda = 0.7$ and $s_0 = 1$. The score is not treated as an oracle; it is exported with each node and used to prioritize ambiguous objects for later inspection.

3.6. Graph Fusion

Candidate nodes are fused by building an association graph over lifted proposals. We disallow same-view association edges by default, so a component represents cross-view object support rather than grouping proposals from a single image. An edge connects two candidates when their centroids are close in 3D and, when valid descriptors exist, their cosine similarity exceeds a feature threshold:

$$\|\mu_a - \mu_b\|_2 \leq \epsilon_g \quad \text{and} \quad \cos(f_a, f_b) \geq \epsilon_f. \quad (4)$$

The feature condition is omitted for candidates without valid descriptors. Connected components define fused object identities. For each component, points are pooled and sampled, descriptors are averaged and renormalized, and the uncertainty is averaged then reduced according to the number of supporting views. This separates view-local evidence from object identity while preserving the proposal provenance needed for debugging and annotation.

Table 1: TUM RGB-D sequences prepared for sparse-view benchmarking. Each sequence contains 10 selected RGB frames and is evaluated with 3, 5, 8, and 10 input views.

Scene ID	Source sequence	Frames	View counts
tum_rgbd_freiburg1_room	freiburg1_room	10	3/5/8/10
tum_rgbd_freiburg1_desk	freiburg1_desk	10	3/5/8/10
tum_rgbd_freiburg1_desk2	freiburg1_desk2	10	3/5/8/10
tum_rgbd_freiburg2_xyz	freiburg2_xyz	10	3/5/8/10
tum_rgbd_freiburg3_long_office_household	freiburg3_long_office_household	10	3/5/8/10

3.7. Relation Inference and Labeling

Spatial relations are inferred from fused centroids using distance and relative coordinate rules. The relation vocabulary includes *near*, *left_of*, *right_of*, *above*, *below*, *in_front_of*, and *behind*. A near relation is emitted for object pairs within a distance threshold; otherwise the vertical displacement or dominant horizontal coordinate determines the directional predicate, with an optional maximum relation distance limiting very distant pairs. For language grounding, fused nodes are labeled by comparing averaged visual descriptors against CLIP text embeddings from an open-vocabulary label set [7]. The exported graph preserves both symbolic fields and numeric support data, allowing the same object identities, point sets, labels, and uncertainty values to be reused by manual review or future learned relation heads.

4. Experiments

4.1. Datasets

We evaluate on public TUM RGB-D indoor sequences [10]. The **freiburg1_room** sequence is used as a development validation scene, and the main benchmark uses the five sequences listed in Table 1. Each sequence contributes 10 selected RGB frames. Although the source benchmark includes depth, our pipeline consumes only RGB frames; the RGB-D benchmark is used because it provides public, recognizable indoor scenes and reproducible sequence identifiers. For every scene, we store the selected view lists, VGGT geometry outputs, SAM proposal files, CLIP+DINOv2 descriptors, graph outputs, generated figures, and checked prediction-seeded object and relation references.

4.2. Sparse-View Protocol

For each benchmark scene, we evaluate 3, 5, 8, and 10 input views. The benchmark runner creates one output directory per scene and view count,

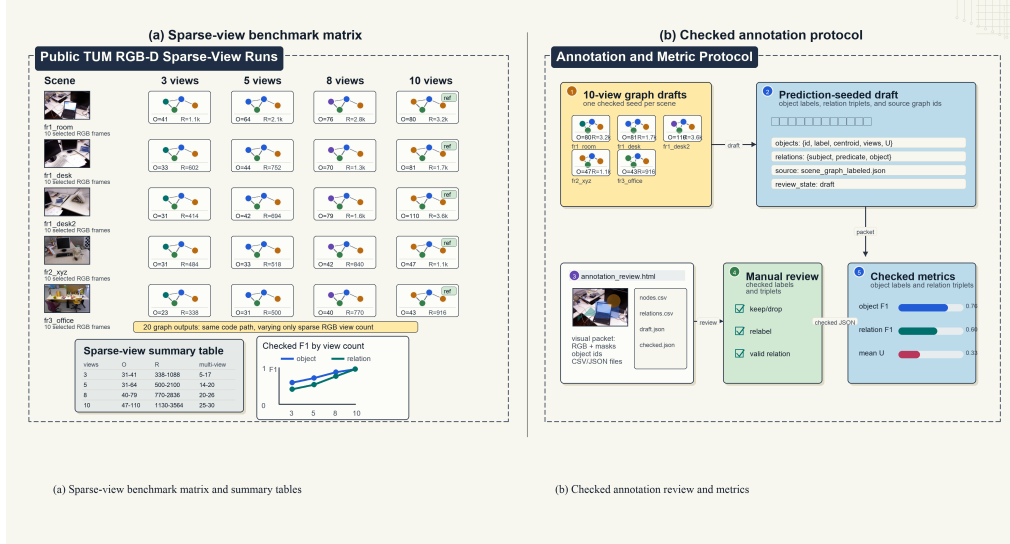


Figure 2: Benchmark and annotation protocol. Panel (a) lays out the full five-scene by four-view-count matrix, with each cell representing one graph output and its object and relation counts; these 20 outputs feed the sparse-view table and view-count metric curve. Panel (b) shows how each 10-view graph becomes a prediction-seeded draft, annotation review packet, checked CSV/JSON files, reviewed label and relation decisions, and reference-consistency object-label and relation-triplet metrics.

then executes geometry extraction, proposal generation, descriptor extraction, graph construction, open-vocabulary labeling, metric export, and figure generation. The same code path is used for single-scene validation and multi-scene experiments. Figure 2 shows two complementary evaluation paths: structural summaries from all 20 sparse-view runs, and a review path that turns the 10-view predictions into checked references for object-label and relation-triplet consistency.

4.3. Comparison Scope

The main benchmark reports the full VGGT+SAM+CLIP+DINOv2 pipeline under a controlled sparse-view protocol. We also include a local component ablation on the development scene to separate the effects of proposal source and descriptor choice. Direct external comparison is nontrivial because many open-vocabulary 3D scene graph systems assume posed RGB-D streams, dense point clouds, or reconstructed maps. We therefore position the benchmark as a reproducible sparse-RGB graph construction baseline and report the input assumptions explicitly.

Table 2: Sparse-view curve on TUM RGB-D `freiburg1_room`.

Views	Candidates	Objects	Multi-view	Relations	Uncertainty	MV Ratio
3	58	27	18	466	0.014525	0.666667
5	98	32	24	628	0.014585	0.750000
8	158	30	26	580	0.011999	0.866667
10	198	32	28	636	0.011513	0.875000

4.4. Metrics

For every run, we report structural and reliability-oriented summaries: object count, relation count, multi-view object count, multi-view ratio, mean uncertainty, and relation density. For the five-scene benchmark, we also report object-label precision, recall, and F1, plus labeled relation-triplet precision, recall, and F1 against checked prediction-seeded reference sets. Because each reference set is initialized from the 10-view graph for the same scene, these scores measure checked sparse-view recovery consistency rather than independent full-scene accuracy. The 10-view row is therefore a reference construction check; the substantive comparison is the 3/5/8-view recovery curve against the checked 10-view reference.

5. Results

5.1. TUM RGB-D Validation

Table 2 summarizes the 3/5/8/10-view validation run on `freiburg1_room`. The run exercises the complete path from RGB frames to VGGT geometry, SAM proposals, CLIP+DINOv2 descriptors, object fusion, open-vocabulary labels, and spatial relations. As the view count increases, lifted proposals increase from 58 to 198, multi-view object support increases from 18 to 28 nodes, and mean uncertainty decreases from 0.014525 to 0.011513. This establishes the expected single-scene behavior before moving to the five-sequence benchmark.

5.2. Component Ablation

Table 3 compares local variants on the development scene. With the number of lifted candidates fixed, moving from geometry-only OpenCV proposals to feature-aware fusion lowers mean uncertainty and increases multi-view support. Replacing OpenCV proposals with SAM masks produces more

Table 3: Development-scene component ablation on a four-view example.

Variant	Proposals	Features	Candidates	Objects	Multi-view	Relations	Uncertainty
OpenCV geometry-only	OpenCV	none	80	14	9	100	0.039071
OpenCV feature fusion	OpenCV	clip+handcrafted	80	16	10	142	0.024012
SAM semantic fusion	SAM	clip	80	22	15	278	0.021765
Full pipeline	SAM	clip+dinov2	80	33	17	618	0.021568

Table 4: Average sparse-view scene graph statistics over five TUM RGB-D sequences.

Views	Scenes	Candidates	Objects	Multi-view	Relations	Uncertainty
3	5	54.00	31.80	11.80	585.20	0.026421
5	5	88.60	42.80	16.80	912.80	0.035933
8	5	140.20	61.40	24.20	1472.40	0.036613
10	5	173.60	72.20	28.20	2104.00	0.034249

fused objects and relations, and adding DINOv2 to CLIP yields the richest graph among the tested variants. This ablation is local rather than a full benchmark, but it supports the design choice of combining class-agnostic mask proposals with complementary language-aligned and self-supervised features.

5.3. Benchmark Results

Table 4 reports the five-sequence sparse-view benchmark. It contains 20 runs: five scenes evaluated at 3, 5, 8, and 10 input views with the same VGGT, SAM, CLIP+DINOv2, labeling, fusion, and relation inference code. Averaged over scenes, increasing the view count from 3 to 10 raises lifted object candidates from 54.0 to 173.6, fused object nodes from 31.8 to 72.2, multi-view object nodes from 11.8 to 28.2, and inferred spatial relations from 585.2 to 2104.0. The growth in multi-view nodes is important: additional views do not merely add isolated proposals, but increase the number of objects supported by evidence from multiple images. The structural counts measure graph coverage; label and relation consistency are evaluated separately below.

Table 5 gives the per-scene breakdown. The full run outputs are stored under `results/benchmark_tum_rgbd_paper_subset/`; aggregate statistics are exported to `sparse_view_scene_graph_summary.csv` and proxy metrics to `sparse_view_scene_graph_metrics.csv`. Checked reference-consistency metrics are exported to `sparse_view_checked_metrics.csv`.

Table 5: Per-scene sparse-view scene graph statistics on the TUM RGB-D benchmark subset.

Scene	Views	Candidates	Objects	Multi-view	Relations	Uncertainty
freiburg1_desk	3	56	33	13	602	0.032715
freiburg1_desk	5	93	44	16	752	0.031488
freiburg1_desk	8	153	70	26	1314	0.037485
freiburg1_desk	10	193	81	30	1718	0.044303
freiburg1_desk2	3	41	31	5	414	0.031997
freiburg1_desk2	5	66	42	14	694	0.040113
freiburg1_desk2	8	113	79	20	1602	0.042473
freiburg1_desk2	10	151	110	25	3564	0.038729
freiburg1_room	3	57	41	10	1088	0.022612
freiburg1_room	5	93	64	15	2100	0.024899
freiburg1_room	8	134	76	25	2836	0.022970
freiburg1_room	10	158	80	30	3192	0.021626
freiburg2_xyz	3	56	31	14	484	0.022882
freiburg2_xyz	5	91	33	19	518	0.022266
freiburg2_xyz	8	144	42	24	840	0.025731
freiburg2_xyz	10	169	47	26	1130	0.024507
freiburg3_long_office_household	3	60	23	17	338	0.021901
freiburg3_long_office_household	5	100	31	20	500	0.060900
freiburg3_long_office_household	8	157	40	26	770	0.054405
freiburg3_long_office_household	10	197	43	30	916	0.042079

Figure 3 shows representative qualitative outputs from the benchmark. The examples show sparse RGB observations converted into fused 3D object nodes, uncertainty-colored graph visualizations, and relation overlays. They also illustrate why uncertainty is retained in the exported graph: sparse-view nodes and edges can remain ambiguous even when the overall graph is coherent, and the uncertainty field provides a practical signal for review.

5.4. Checked Reference-Consistency Metrics

For the five-scene benchmark, each 10-view graph is converted into a prediction-seeded annotation draft and checked through the review CSV/JSON path. We compare the 3/5/8-view outputs against those checked reference sets and include the 10-view row as a reference construction check. Table 6 summarizes the five-scene average. Object-label F1 improves from 0.622 at 3 views to 0.751 at 5 views and 0.914 at 8 views. Labeled relation-triplet F1 improves from 0.434 to 0.561 and 0.799 over the same view counts. Precision is high even at 3 views, while recall rises substantially with more views; the main sparse-view failure mode is therefore missing objects and relation triplets rather than producing many inconsistent labels.

Table 6: Average checked reference-consistency results over five TUM RGB-D sequences. The 10-view row checks the prediction-seeded reference construction rather than independent accuracy.

Views	Scenes	Obj P	Obj R	Obj F1	Rel P	Rel R	Rel F1
3	5	0.978	0.467	0.622	0.923	0.293	0.434
5	5	0.987	0.620	0.751	0.920	0.420	0.561
8	5	0.984	0.857	0.914	0.955	0.701	0.799
10	5	1.000	1.000	1.000	1.000	1.000	1.000

Table 7 gives the per-scene consistency scores. The hardest case is **freiburg1_desk2**, where relation-triplet F1 is 0.206 at 3 views and 0.616 at 8 views, reflecting the difficulty of recovering a dense relation set from sparse inputs. The strongest 8-view relation score is on **freiburg1_room** at 0.929, which is also the scene used during development. This spread shows that the aggregate trend is not caused by one sequence while keeping the failure modes visible.

6. Discussion

The experiments support the central premise of this paper: feed-forward geometry and 2D foundation-model semantics can be connected into an explicit sparse-view 3D scene graph without optimizing a complete scene representation. The five-sequence benchmark shows that the same implementation runs across several indoor sequences and produces more object and relation evidence as additional views are provided. The checked reference-consistency curves further show that the graph outputs become closer to the 10-view references in both object labels and relation triplets, with the strongest gain coming from improved recall.

The main limitation is evaluation scope. The reported F1 values are measured against checked annotations initialized from the 10-view prediction, so they measure sparse-view recovery relative to a higher-view graph rather than absolute recognition against independently authored dense-scene labels. The 10-view row is therefore a reference construction check, not a separate performance result. A second limitation is relation expressivity: the current predicates are inferred from centroid and distance rules. These rules are inspectable and easy to audit, but they cannot capture the full range of

Table 7: Per-scene consistency results against checked prediction-seeded references. The 10-view rows check reference construction rather than independent accuracy.

Scene	Views	Obj P	Obj R	Obj F1	Rel P	Rel R	Rel F1
freiburg1_desk	3	1.000	0.407	0.579	0.987	0.346	0.512
freiburg1_desk	5	1.000	0.543	0.704	0.992	0.434	0.604
freiburg1_desk	8	0.971	0.840	0.901	0.962	0.736	0.834
freiburg1_desk	10	1.000	1.000	1.000	1.000	1.000	1.000
freiburg1_desk2	3	1.000	0.282	0.440	0.990	0.115	0.206
freiburg1_desk2	5	1.000	0.382	0.553	0.954	0.186	0.311
freiburg1_desk2	8	1.000	0.718	0.836	0.993	0.446	0.616
freiburg1_desk2	10	1.000	1.000	1.000	1.000	1.000	1.000
freiburg1_room	3	0.976	0.500	0.661	0.961	0.328	0.489
freiburg1_room	5	1.000	0.800	0.889	0.986	0.648	0.782
freiburg1_room	8	1.000	0.950	0.974	0.987	0.877	0.929
freiburg1_room	10	1.000	1.000	1.000	1.000	1.000	1.000
freiburg2_xyz	3	1.000	0.660	0.795	0.955	0.409	0.572
freiburg2_xyz	5	1.000	0.702	0.825	0.903	0.414	0.568
freiburg2_xyz	8	1.000	0.894	0.944	0.971	0.722	0.828
freiburg2_xyz	10	1.000	1.000	1.000	1.000	1.000	1.000
freiburg3_long_office_household	3	0.913	0.488	0.636	0.722	0.266	0.389
freiburg3_long_office_household	5	0.935	0.674	0.784	0.764	0.417	0.540
freiburg3_long_office_household	8	0.950	0.884	0.916	0.862	0.725	0.788
freiburg3_long_office_household	10	1.000	1.000	1.000	1.000	1.000	1.000

support, containment, functional, or activity relations studied in richer 3D scene graph settings [19, 20, 9].

These limitations point to a clear next stage. The generated review packets expose RGB thumbnails, projected object crops, object labels, relation triplets, and uncertainty values for each scene. Expanding this review from prediction-seeded references to independently authored annotations would turn the current consistency analysis into standard object-label and relation-triplet evaluation. The same annotations would also support a learned relation module, allowing the present rule-based predicates to be replaced once sufficient object-relation supervision is available.

7. Conclusion

We presented an uncertainty-aware sparse-view 3D scene graph pipeline that combines feed-forward VGGT geometry with SAM proposals, CLIP+DINOv2 descriptors, object fusion, open-vocabulary labeling, and spatial relation inference. The accompanying benchmark tooling turns

public RGB-D sequences into graph outputs, paper tables, qualitative figures, checked sparse-view reference scores, and review packets. The result is a reproducible baseline for moving from sparse RGB captures to inspectable language-grounded 3D object-relation graphs.

Data Availability

The RGB-D source sequences used in this study are publicly available from the TUM RGB-D benchmark. The selected sparse-view manifests, generated scene-graph summaries, annotation review packets, checked annotation JSON files, and metric tables are provided as supplementary research artifacts. Model checkpoints and third-party components follow their respective licenses.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

OpenAI ChatGPT and OpenAI Codex were used for code inspection, experiment bookkeeping, manuscript organization, and language editing. The author reviewed all AI-assisted output and takes full responsibility for the article. Generative AI was not used to create or alter experimental images, metrics, or figures.

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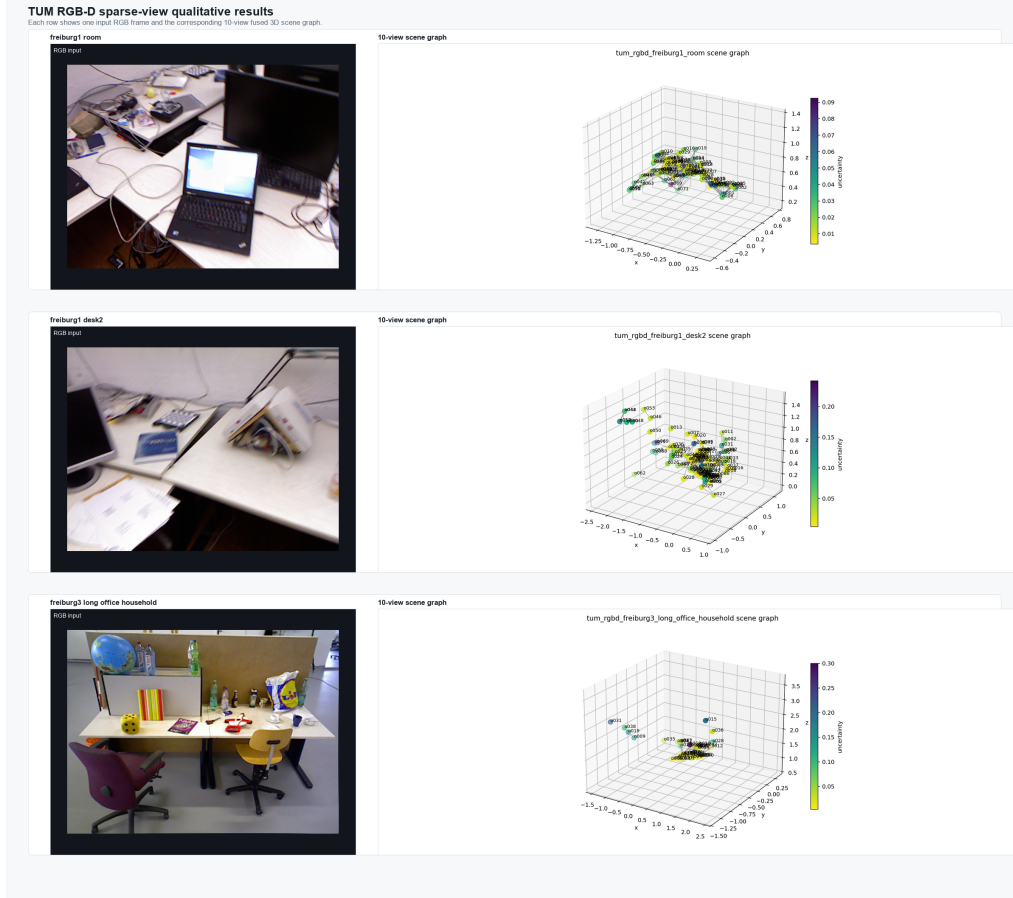


Figure 3: Qualitative examples from the TUM RGB-D benchmark subset. Each row shows one input RGB frame and the corresponding 10-view fused 3D scene graph visualization. Node colors encode uncertainty and edge overlays show inferred near relations.